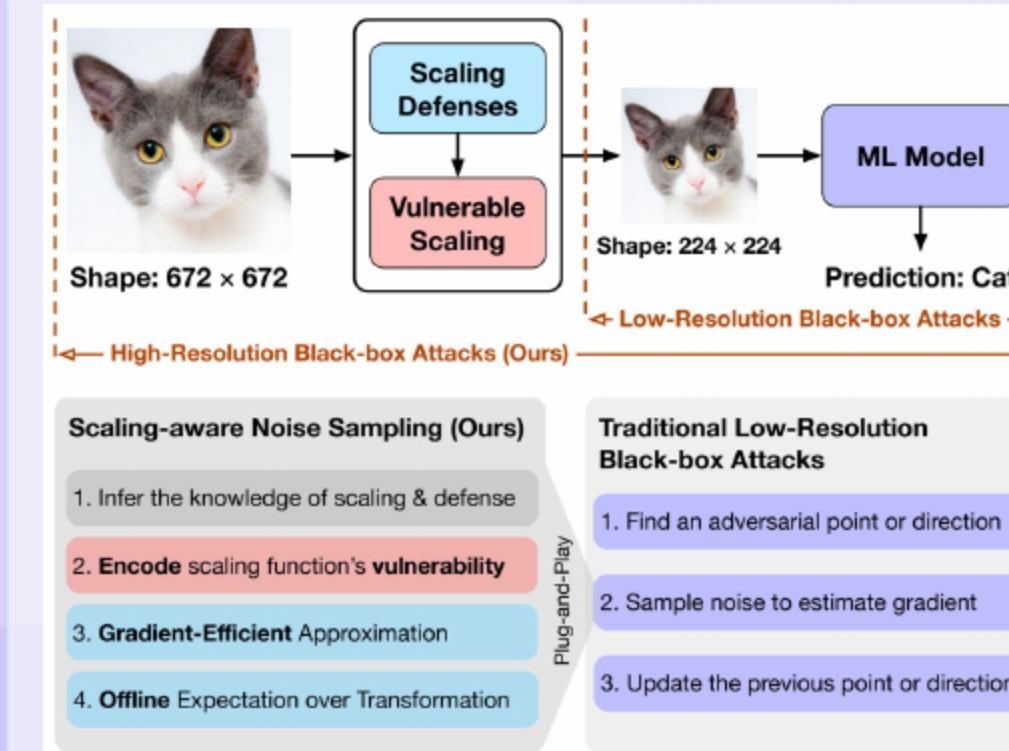


Rethinking Image-Scaling Attacks: The Interplay Between Vulnerabilities in Machine Learning Systems

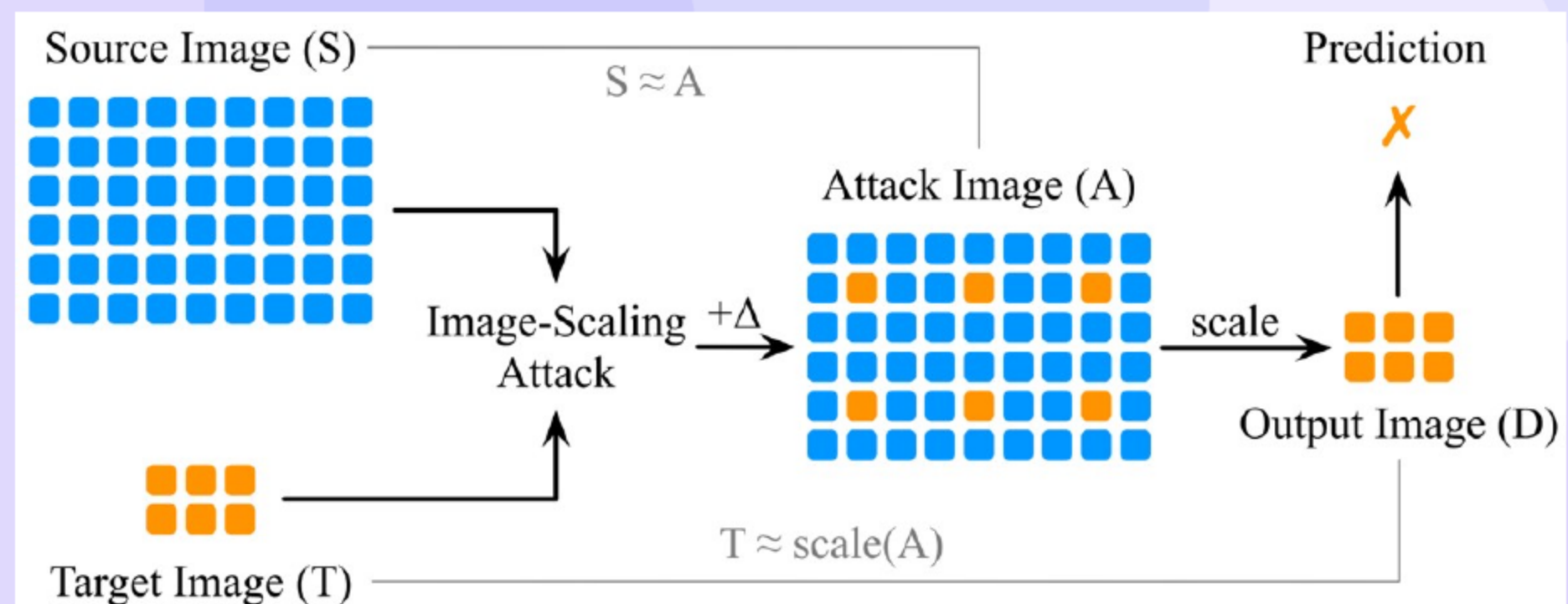
Yue Gao; Ilia Shumailov; Kassem Fawaz

01 Pipeline-aware black-box attacks: motivation and contributions



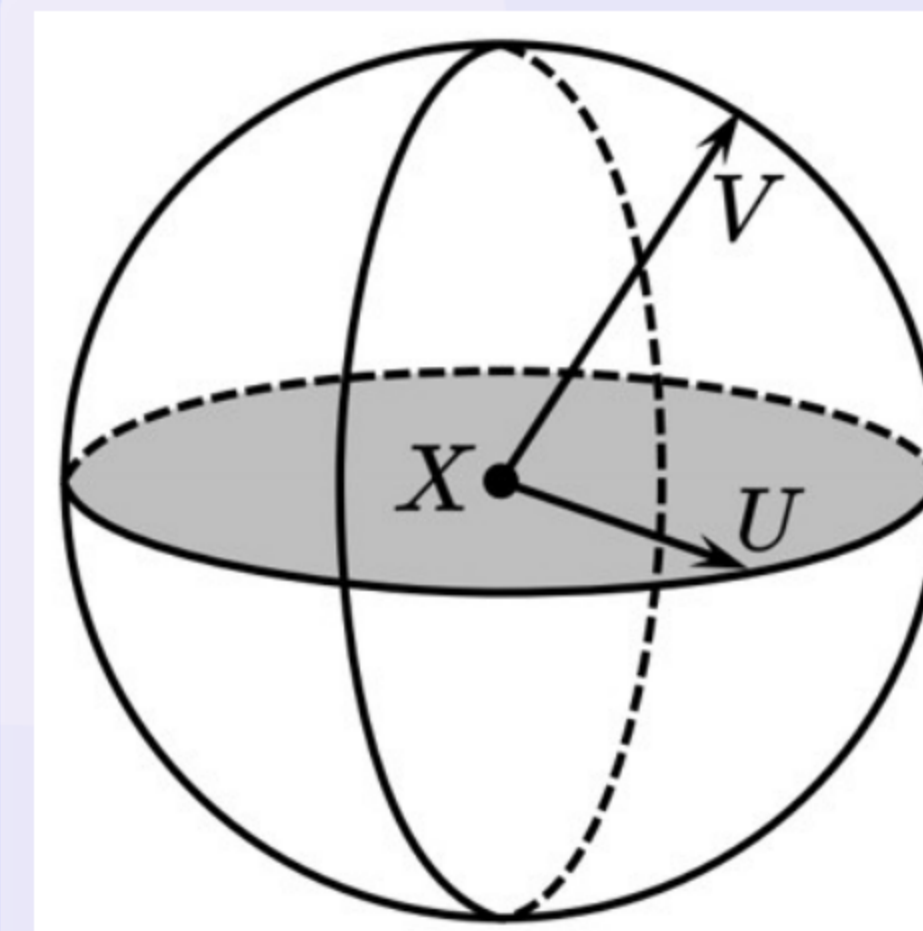
- End-to-end attacks; introduce \SNS for hidden perturbations.
- SNS plugs into HSJ/Sign-OPT; better gradients; offline EOT.
- 4/5 defenses exploitable; pipeline-aware attacks \beat model-only.

02 Pipeline and goal + mechanism/defenses overview



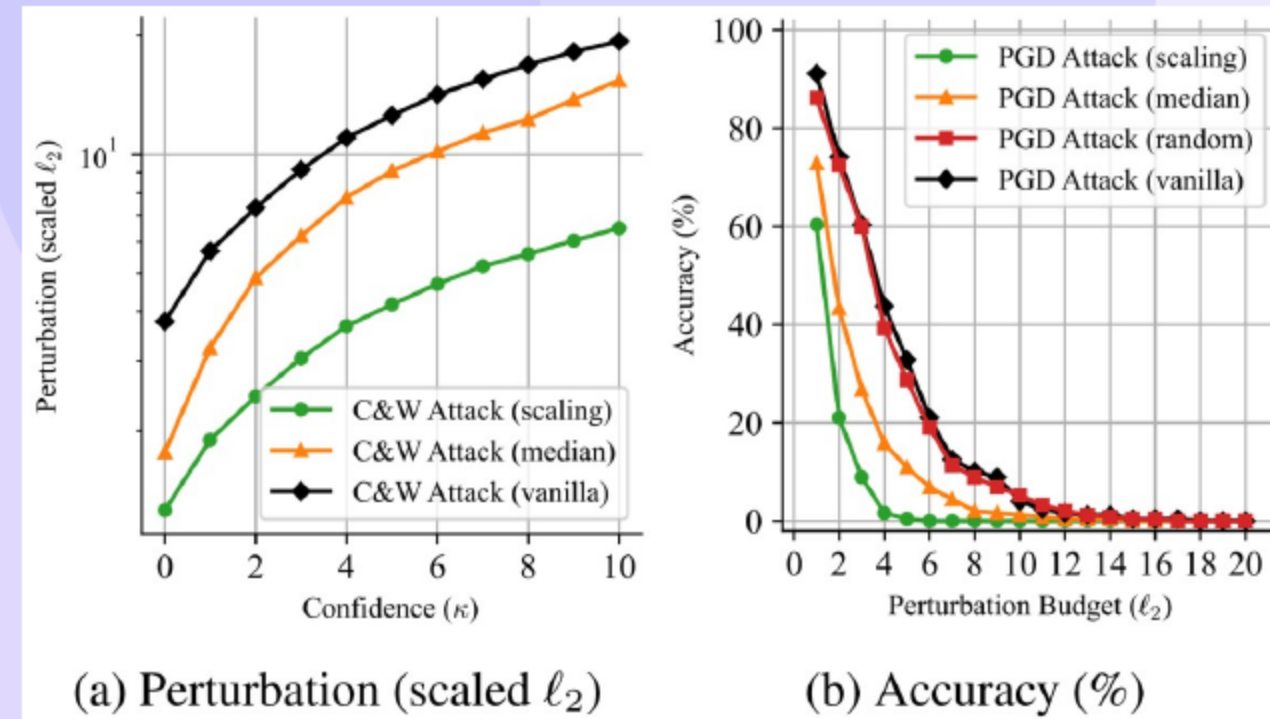
- Pipelines preprocess and downscale before f.
- Decision-only; infer $h \circ g$ once via brute force.
- Defenses: median/random; detectors: unscale, min-filter, spectrum.

03 SNS and plug-in with HSJ/Sign-OPT



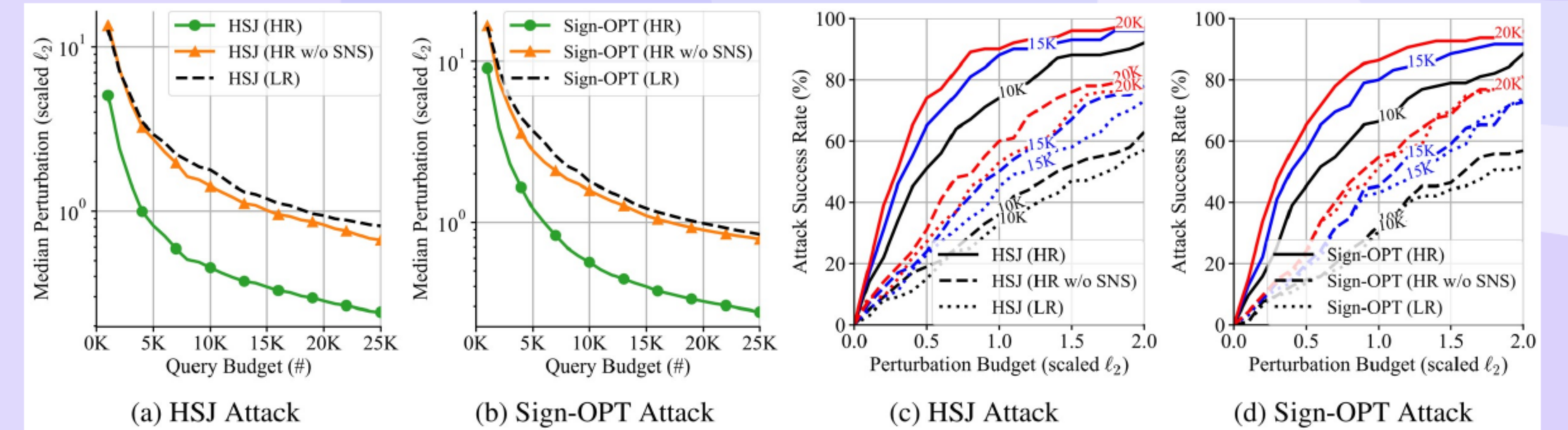
- Decision attacks loop: boundary search, noise gradient, update.
- SNS samples in LR and projects to HR to hide perturbations.
- Efficient projection uses objective gradients; fast, sufficient estimates.
- SNS plugs into HSJ and Sign-OPT to form high-resolution variants.

04 Handling defenses: median vs. randomized + offline EOT



- Median filtering blunts line search; trimmed, weighted averaging restores signal.
- Offline EOT replaces $h \circ g$ in SNS; no extra queries.
- Randomized filtering averages to uniform, removing the hiding channel.
- Closed-form expectations stabilize some defenses.
- Left: HR needs less scaled L2; Right: accuracy drops slower under randomized defense.

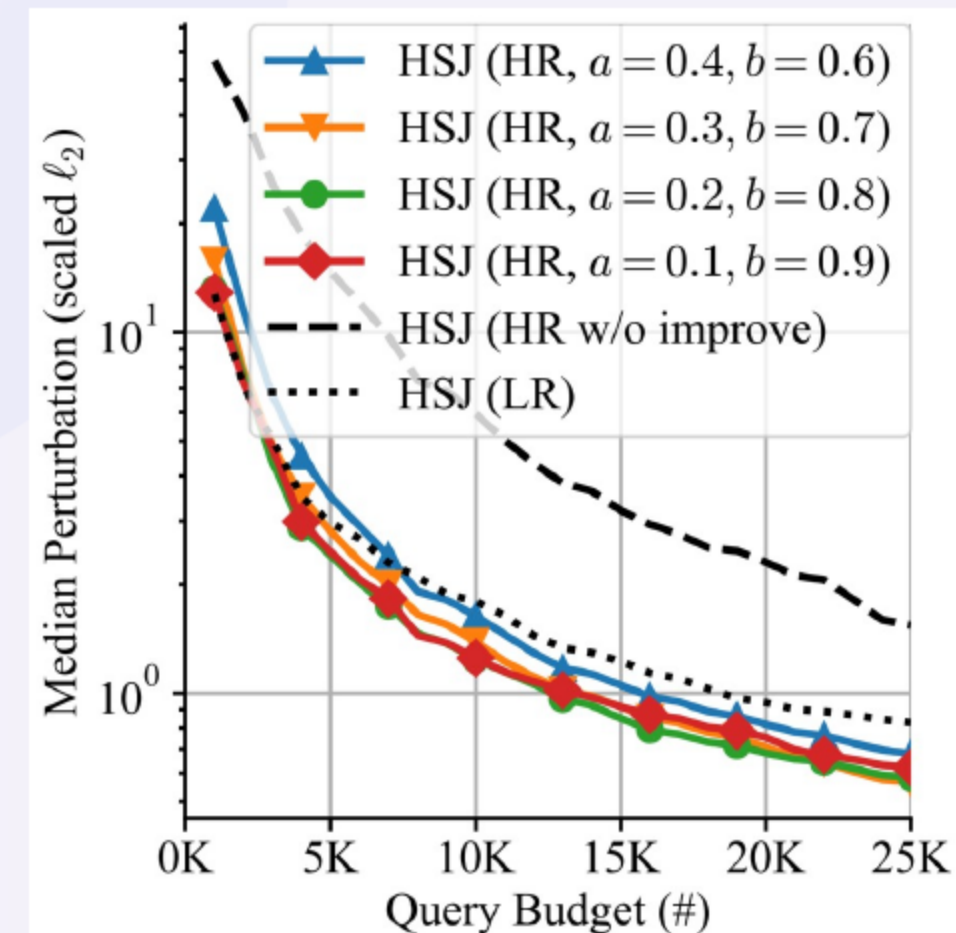
05 Setup, metrics, and key questions



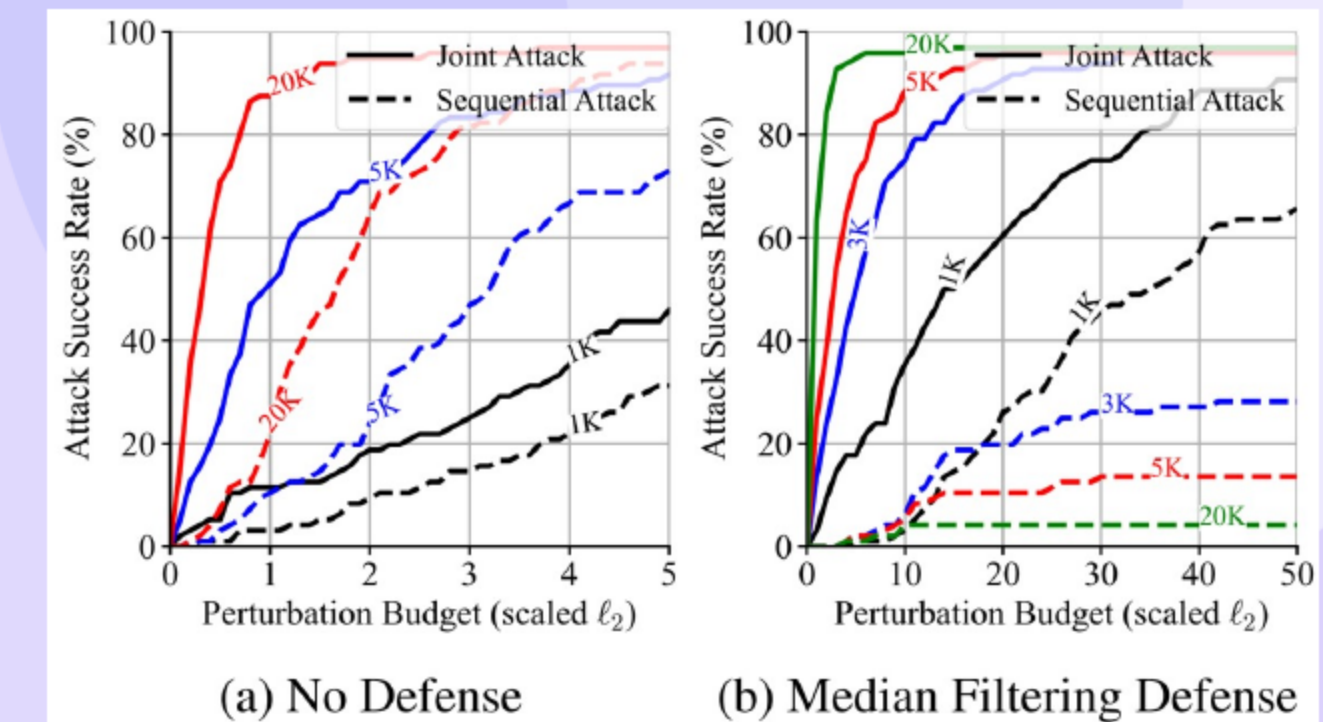
- Experiments: ImageNet/ResNet-50; CelebA/ResNet-34; OpenCV linear scaling.
- ASR: fraction with $F(X') \neq F(X)$ within budgets.
- Scaled L2: $\|g(X') - g(X)\|_2$ after scaling (primary).
- Query: one call to F ; offline EOT adds zero.
- Key questions: which defenses fail; when do HR attacks help?

06 Results and defense deep-dive (median filtering)

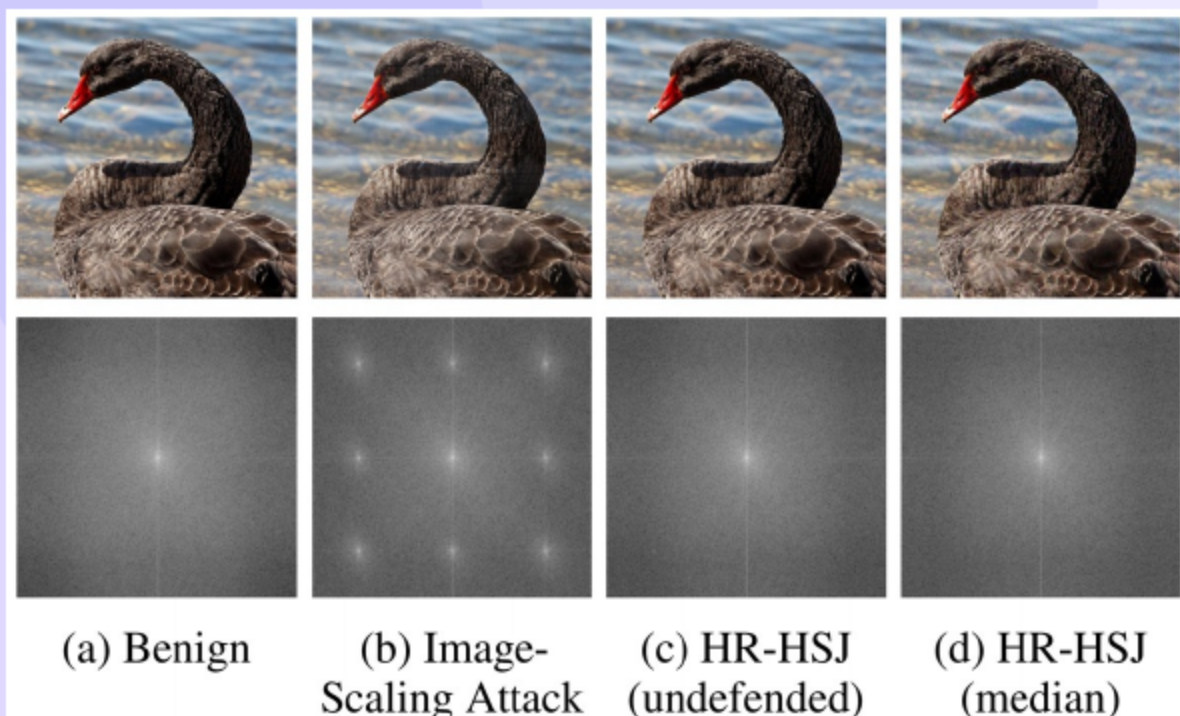
- Undefended: HR has lower scaled L2 and higher ASR than LR.
- Efficient SNS matches precise projection at much lower compute.
- Median filtering: improved estimation speeds convergence and preserves HR gains.
- Median defense needs many flips in narrow ranges; structure remains.
- Disabling SNS degrades performance and narrows the HR-LR gap.
- Curve gap shows HR beats LR under median filtering.



07 Ablations + Cloud API



- Joint attack over g and f beats sequential.
- Sequential degrades with more queries; joint maintains gains.
- Decision-based API: 100 ImageNet, 3K queries each.
- \$1.18/image; HR > LR at equal budgets.
- Joint > sequential across budgets; HR > LR on API.



- Robust downscalers/preprocessors that remove hiding channels without hurting utility.
 - Pipeline-aware detectors with low false positives.
 - Beyond post-scaling L2 and decision-only: new norms and attacks.
- Transfer across scaling ratios, interpolation, and randomness.
 - Efficiently infer unknown $h \circ g$ in the wild.
 - Hidden LR appears after unscaling/min-filter; median reduces visibility.

Thank You

Questions & Discussion